

Computability: Exercise 5

Due: May 11, 2026

Question 1

Define

$$\text{SELF-NO-ACC}_{\text{TM}} \stackrel{\text{def}}{=} \{ \langle M \rangle : M \text{ does not accept } \langle M \rangle \}.$$

Use a direct argument (without using reductions) to prove that $\text{SELF-NO-ACC}_{\text{TM}} \notin \text{R}$.

Hint: Repeat arguments similar to those in the proof that $\text{HALT}_{\text{TM}} \notin \text{R}$.

Question 2

- Prove that for every $L \in \text{R}$, $L^* \in \text{R}$.
- Prove that RE is closed under concatenation, that is, if $L_1, L_2 \in \text{RE}$ then $L_1 \cdot L_2 \in \text{RE}$.
- Prove that for every $L \in \text{RE}$, $L^* \in \text{RE}$.

Question 3

In class we showed how one can translate a Turing Machine M to a TM M' such that if M stops on w then $M'(w) = M(w)\#t$, where t is the length of the run of M on an input w .

- Assume that M stops on w and runs for t steps on it. Show that if $t > |w|$, the running time of M' on w is bounded by $O(t^2)$.
- Suggest a way to tweak M' so that if M stops on w and runs for t steps, then the running time of M' is bounded by $O(t \log t)$ (We still assume that $t > |w|$).

Question 4

In class we discussed the implementation of a universal Turing Machine U , in which U repeatedly transforms the tape from containing $\langle M \rangle \langle c \rangle$ to containing $\langle M \rangle \langle c^+ \rangle$, where c is a configuration of $\langle M \rangle$ and c^+ is its successive configuration.

- Explain how one can implement U in a way such that it performs each such transformation in time that is polynomial in $|\langle M \rangle \langle c \rangle|$, namely in the length of the contents of the tape.
- Can you think of a way to change U such that each computation of the next configuration from the previous one takes time that is only polynomial in $|\langle M \rangle|$?

Hint: the encoding of $\langle M \rangle$ will no longer stay at the leftmost part of the tape, and one may assume that all the encodings of letters and states of M are of the same length.

Question 5

Prove the following parts of the reduction theorem, that is, supposing $L_1 \leq_m L_2$:

- a. If $L_2 \in \text{RE}$, then $L_1 \in \text{RE}$.
- b. If $L_2 \in \text{R}$, then $L_1 \in \text{R}$.

Question 6

Prove the following, assuming that all the languages below are non-trivial, that is, they are not empty nor is their complement.

- a. If $L_1 \leq_m L_2$ and $L_2 \leq_m L_3$, then $L_1 \leq_m L_3$.
- b. If $L_1 \leq_m L_2$ then $\overline{L_1} \leq_m \overline{L_2}$.
- c. If $L, \overline{L} \in \text{RE}$ then $L \in \text{R}$.
- d. If $L \in \text{RE}$ and $L \leq_m \overline{L}$ then $L \in \text{R}$.
- e. $L \in \text{R}$ if and only if for every non-trivial language L' , $L \leq_m L'$.
- f. There is a pair of non-trivial languages L_1, L_2 such that $L_1 \leq_m L_2$ but $L_2 \not\leq_m L_1$.